

Ghofrane TRIKI

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Profile

Computer Engineering graduate (IIT, class of 2026 – Highest Honors) specializing in Software Engineering, Business Intelligence, and Information Systems Security. Experienced in full-stack web and mobile development, multi-tenant SaaS architecture, DevOps, data science, machine learning, and cybersecurity fundamentals. Seeking a software engineering role to build secure, scalable systems and contribute to innovative projects in a stimulating professional environment.

Education

Computer Engineering Degree – *Highest Honors* *Sep 2023 – Jun 2026*
International Institute of Technology (IIT), Sfax, Tunisia

- Major: Software Engineering and Business Intelligence (GLID)
- Minor: Information Systems Security (Cybersecurity)
- Capstone Project (PFE): DigiMart – Multi-tenant SaaS e-commerce platform (Highest Honors)
- Key coursework: Secure Software Design, Network Security, Cryptography, OWASP, Data Mining, OLAP, Machine Learning, Distributed Systems, Clean Architecture, Agile/Scrum

Preparatory Cycle – Mathematics and Physics *Sep 2021 – Jun 2023*
Preparatory Institute for Engineering Studies of Sfax (IPEIS), Tunisia

- Intensive preparation in Mathematics, Physics and Engineering Sciences

Baccalaureate in Experimental Sciences – With Honors *Jun 2021*
Habib Thameur High School, Sfax, Tunisia

Professional Experience

Full-Stack Software Engineer – Capstone Project (PFE) *Jan 2026 – Jun 2026*
DigiLab Solutions, Sfax, Tunisia

- Designed and developed **DigiMart**, a production-grade multi-tenant SaaS e-commerce platform enabling independent merchants to manage online stores and physical point-of-sale from a single unified system
- Backend built on **Clean Architecture** (4 Maven modules: domain, application, infrastructure, api) with Spring Boot 4 and Java 21; React 19 / Vite 7 frontend organized using Atomic Design
- Implemented **security-first multi-tenancy**: strict per-store data isolation, opaque Bearer token authentication with SHA-256 hashing, granular RBAC with role templates, and exportable audit logs
- Applied OWASP secure coding practices: input validation, parameterized queries, least-privilege access control, HTTPS enforcement, and rate limiting via NGINX
- Delivered 19 business modules: catalog, product variants, multi-warehouse stock, orders, CRM, promotions, loyalty, POS terminal, public storefront, theme engine, wallet, subscriptions, and DevOps pipeline
- Authored **133 automated tests** (JUnit, Testcontainers, ArchUnit, Vitest, Playwright) – 100% passing; architecture fitness functions via ArchUnit preventing layer coupling violations
- Built CI/CD pipeline with GitHub Actions: build, test, Docker image push to GHCR, Docker Compose deployment on Ubuntu VPS (OVH), NGINX reverse proxy, automatic rollback on failure
- Technologies: Java 21, Spring Boot 4, React 19, Vite 7, PostgreSQL 15, Flyway, Docker, NGINX, GitHub Actions, GHCR, JUnit, Testcontainers, ArchUnit, Vitest, Playwright, Gemini API

Mobile Developer Intern *Feb 2025 – Jun 2025*
Exadev Company, Sfax, Tunisia

- Developed a full-stack mobile application for gym management using React Native and Laravel
- Designed and managed a MySQL database covering classes, training programs, and user accounts
- Built RESTful endpoints with Laravel and integrated them with the React Native mobile frontend

- Technologies: React Native, React, Laravel, MySQL, REST API

Full-Stack Developer Intern – End-of-Year Project (PFA)

Feb 2025 – Jun 2025

Eventizer, Sfax, Tunisia

- Developed NexiQuest, an online education platform featuring interactive learning modules and content management
- Technologies: Laravel, Blade, MySQL

Web Developer Intern

Feb 2024 – Jun 2024

Exadev Company, Sfax, Tunisia

- Built Fimochic, a full-featured e-commerce website with product catalog, sales, and inventory management
- Integrated advanced JavaScript interactions and interactive AR filters via Meta Spark Studio
- Technologies: Laravel, Blade, MySQL, JavaScript, Meta Spark Studio

Academic Projects

Full-Stack Web Application – Spring Boot / Angular (Training Center Management)

2025 – 2026

- Built a training center management system with layered backend architecture and Angular frontend supporting SSR and CSR rendering modes
- Technologies: Spring Boot, Angular, REST API

FimoChic – Business Intelligence & Decision Support System

2024 – 2025

- Designed an end-to-end BI solution to analyze e-commerce sales performance and support strategic decision-making
- Built ETL pipelines with SSIS to extract, transform, and load data from operational sources into a data warehouse
- Developed multidimensional OLAP cubes with SSAS enabling slice-and-dice analysis across time, product, and customer dimensions
- Created interactive Power BI dashboards for KPI visualization (revenue trends, conversion rates, top products)
- Technologies: SQL Server, SSIS, SSAS, Power BI

Business Process Modeling – Odoo ERP

2024 – 2025

- Modeled purchasing and sales business processes using BPMN notation within an ERP system
- Configured and customized Odoo modules to match real-world business requirements; automated operational workflows
- Technologies: Odoo ERP

Insurance Claims Prediction – Data Mining & Machine Learning

2024 – 2025

- Performed predictive analysis on insurance datasets to forecast claims and optimize policy parameters
- Applied data preprocessing, feature engineering, and model selection across classification and regression algorithms
- Evaluated model performance using cross-validation, ROC and AUC metrics
- Technologies: Python, Pandas, Scikit-learn, Machine Learning, Data Mining

FashionAcademy – Web Management Platform

2024 – 2025

- Developed a complete platform for managing a fashion academy: members, trainees, designers, collections, and fashion show scheduling
- Integrated Firebase Authentication for secure login and Firestore for real-time data storage and synchronization
- Technologies: Angular, Firebase Authentication, Firestore, JSON Server

Fimochic WebApp – Full-Stack Redesign

2024 – 2025

- Complete architectural redesign of the Fimochic e-commerce platform with a modern decoupled stack
- Built a high-performance responsive React frontend and a robust REST API backend with .NET Core

- Technologies: React, .NET Core, REST API

QuizApp – Cross-Platform Mobile Application

2024 – 2025

- Built an interactive cross-platform mobile quiz app with dynamic question categories and score tracking
- Integrated Open Trivia Database REST API for varied, up-to-date question sets
- Technologies: Flutter, REST API

Version Control System – Git-like (C / Linux)

2023 – 2024

- Implemented a version control system inspired by Git in C with snapshot management, SHA-256 content hashing, and a recursive WorkTree for tracking file and directory state
- Designed save and restore operations with linked list data structures and low-level Unix file system manipulation
- Technologies: C, Bash, SHA-256, Linked Lists, Unix File Management

Technical Skills

Languages: Java, Python, JavaScript, PHP, C

Backend: Spring Boot 4, Java 21, Clean Architecture, Laravel, .NET Core, REST API, Maven multi-modules

Frontend: React 19, Vite 7, Angular, HTML5, CSS3, Bootstrap, Atomic Design, SSR, CSR

Mobile: React Native, Flutter

Databases: PostgreSQL, MySQL, Oracle, Firebase Firestore, Flyway

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, NGINX, Ubuntu VPS, GHCR, AWS (Fundamentals), Linux, Bash Scripting

Testing: JUnit, Testcontainers, ArchUnit, Vitest, Playwright, Testing Library, Spring Security Test

Security & Cybersecurity: Spring Security, Opaque Bearer Token, SHA-256, RBAC, OWASP Top 10, Input Validation, Parameterized Queries, Rate Limiting, Audit Logging, Information Systems Security

Business Intelligence & Data Science: SQL Server, SSIS, SSAS, Power BI, ETL Pipelines, OLAP Cubes, Pandas, Scikit-learn, Machine Learning, Data Mining

Tools: Git, GitHub, Postman, Figma, Jira, Odoo ERP, Firebase, Meta Spark Studio

Languages

Arabic: Native • **French:** Fluent – B2 Level • **English:** Fluent – IELTS Preparation • **Italian:** Intermediate

Extra-Academic Activities

Entrepreneurship: Co-founder and manager of Fimochic, a handmade craft e-commerce business (polymer clay jewelry) – handled product design, pricing strategy, online sales, and customer relations

IIT Competitions (2023–2024):

- 1st place – Theater competition “Together on Stage”
- 1st place – Slam poetry competition “The Golden Pen”
- 3rd place – Mathematics competition “The Mathematical Labyrinth”

Personal Skills

Creativity • Problem Solving • Critical Thinking • Leadership • Teamwork • Punctuality

Interests

Creative Programming • Art and Design • Zumba